THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 187	$34(78-111)=n_2(72-78)$.
2	**(a)** $7:9$	Amounts are in the ratio $(41-34):(34-25)=7:9$.
3	**(b)** 32	$\frac{2\times20\times80}{20+80}=32$ (total distance / total time).
4	**(b)** 69	New salary is higher by $n\times2=44$ thousand.
5	**(c)** 100	Sum of the extra numbers $=88\times100-48\times75=5200$; divide by 52.
6	**(c)** 22%	$\frac{2000\times13+9000\times24}{11000}=22\%$.
7	**(c)** 54%	$\frac{750\times34+1250\times66}{2000}=54\%$.
8	**(d)** 60	$\frac{72\times90+144\times45}{216}=60$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{20\times60+30\times70}{50}=\frac{3300}{50}=66$, using R.
10	**(d)** A is false but R is true.	A is false when the groups have different sizes (the weights differ); R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let amounts be x and y; $16x+64y=36(x+y)$ (1); $20x=28y$; $x:y=7:5$ (1).	1+1
12	Take distance each way $=90$ km, so total distance $=180$ km; Time $=\frac{90}{30}+\frac{90}{45}=5$ h (1); Average speed $=\frac{180}{5}=36$ km/h (not the simple mean of 30 and 45) (1).	1+1
13	Rise in total $=29\times2=58$ thousand; New salary $=$ old salary $+$ rise in total (1); $=26+58=84$ thousand, i.e. ₹84000 (1).	1+1
14	Let amounts be x and y; $52x+88y=67(x+y)$ (1); $15x=21y$; $x:y=7:5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $40\times49+90n=70(40+n)$ (1); $n(90-70)=40(70-49)$ (1); $n=42$ (1).	1+1+1
16	Weighted sum $=80\times4+78\times3+54\times8+85\times2=1156$ (1); Sum of weights $=17$ (1); Weighted average $=68$ (1).	1+1+1
17	Original sum $=714$; New sum $=714+68+50=832$ (1); New mean $=\frac{832}{16}=52$ (1); Rises (1).	1+1+1
	OR: Let the number of remaining students be n; $78\times71+80n=74(78+n)$ (1); $n(80-74)=78(74-71)$ (1); $n=39$ (1).	1+1+1
18	Weighted sum $=58\times3+52\times5+51\times2+55\times4=756$ (1); Sum of weights $=14$ (1); Weighted average $=54$ (1).	1+1+1
	OR: Original sum $=819$; New sum $=819+81+60=960$ (1); New mean $=\frac{960}{15}=64$ (1); Rises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $40\times72=2880$ (1). (ii) $60\times82=4920$ (1). (iii) $\frac{2880+4920}{100}=78$ (1). (iv) Section C total $=50\times90=4500$ (1); new average $=\frac{7800+4500}{150}=\frac{12300}{150}=82$ (1).	1+1+1+2
20	Cost of the mixture $=\frac{2\times145+3\times150}{5}$ (1); $=148$ per kg (1); Profit factor $=\frac{125}{100}$ (1); Selling price $=185$ per kg (2).	1+1+1+2
	OR: January food $=1656$ (1); February food $=960$ (1); January is higher by 696 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2

Mistakes to avoid:

- Taking the combined average as $\frac{a_1 + a_2}{2}$ when the groups have different sizes.
- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. 20% of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 36	$\frac{2\times30\times45}{30+45}=36$ (total distance / total time).
2	**(d)** 105	New salary is higher by $n\times3=69$ thousand.
3	**(b)** 19	Sum of the extra numbers $=53\times78-51\times71=513$; divide by 27.
4	**(b)** 20%	$\frac{4000\times32+6000\times12}{10000}=20\%$.
5	**(d)** 46%	$\frac{780\times70+1440\times33}{2220}=46\%$.
6	**(c)** 83	$\frac{80\times93+100\times75}{180}=83$.
7	**(d)** 42	$42(91-110)=n_2(72-91)$.
8	**(c)** $15:23$	Amounts are in the ratio $(64-49):(49-26)=15:23$.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R is a special case and does not explain why the combined average lies between the two averages.
10	**(c)** A is true but R is false.	Pure amount $=20+60=80$ mL in 500 mL, i.e. 16%, so A is true; R is false because the volumes differ.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take distance each way $=308$ km, so total distance $=616$ km; Time $=\frac{308}{44}+\frac{308}{77}=11$ h (1); Average speed $=\frac{616}{11}=56$ km/h (not the simple mean of 44 and 77) (1).	1+1
12	Rise in total $=23\times3=69$ thousand; New salary $=$ old salary $+$ rise in total (1); $=38+69=107$ thousand, i.e. ₹107000 (1).	1+1
13	Let amounts be x and y; $41x+72y=64(x+y)$ (1); $23x=8y$; $x:y=8:23$ (1).	1+1
14	Take distance each way $=90$ km, so total distance $=180$ km; Time $=\frac{90}{30}+\frac{90}{90}=4$ h (1); Average speed $=\frac{180}{4}=45$ km/h (not the simple mean of 30 and 90) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $70\times67+51n=65(70+n)$ (1); $n(51-65)=70(65-67)$ (1); $n=10$ (1).	1+1+1
16	Weighted sum $=52\times6+61\times8+72\times6+82\times4=1560$ (1); Sum of weights $=24$ (1); Weighted average $=65$ (1).	1+1+1
17	Original sum $=448$; New sum $=448+51+29=528$ (1); New mean $=\frac{528}{16}=33$ (1); Rises (1).	1+1+1
	OR: Let the number of remaining students be n; $56\times97+87n=95(56+n)$ (1); $n(87-95)=56(95-97)$ (1); $n=14$ (1).	1+1+1
18	Weighted sum $=55\times6+61\times2+72\times3+53\times6=986$ (1); Sum of weights $=17$ (1); Weighted average $=58$ (1).	1+1+1
	OR: Original sum $=418$; New sum $=418+52+37=507$ (1); New mean $=\frac{507}{13}=39$ (1); Rises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 30% of 4 L $=1.2$ L (1). (ii) 20% of 6 L $=1.2$ L (1). (iii) Pure juice $=2.4$ L in 10 L, so 24% (1). (iv) Let the total volume be V: $0.2V=2.4\Rightarrow V=12$ L (1); water to add $=12-10=2$ L (1).	1+1+1+2
20	Cost of the mixture $=\frac{4\times120+2\times180}{6}$ (1); $=140$ per kg (1); Profit factor $=\frac{120}{100}$ (1); Selling price $=168$ per kg (2).	1+1+1+2
	OR: January food $=1748$ (1); February food $=1800$ (1); February is higher by 52 (1); The larger total outweighs the smaller share in February (2).	1+1+1+2

Mistakes to avoid:

- Taking the combined average as $\frac{a_1 + a_2}{2}$ when the groups have different sizes.
- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. 20% of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 45	Sum of the extra numbers $= 61 \times 105 - 48 \times 80 = 2565$; divide by 57.
2	**(a)** 33%	$\frac{2000 \times 15 + 3000 \times 45}{5000} = 33\%$.
3	**(d)** 12%	$\frac{910 \times 18 + 780 \times 5}{1690} = 12\%$.
4	**(d)** 73	$\frac{50 \times 94 + 150 \times 66}{200} = 73$.
5	**(c)** 112	$20(53 - 81) = n_2(48 - 53)$.
6	**(c)** 17 : 18	Amounts are in the ratio $(89 - 72) : (72 - 54) = 17 : 18$.
7	**(a)** 40	$\frac{2 \times 28 \times 70}{28 + 70} = 40$ (total distance / total time).
8	**(c)** 100	New salary is higher by $n \times 4 = 76$ thousand.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Every bar represents 100%, so all bars have equal length.
10	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{600 + 1200}{5} = 360$, using R.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rise in total $= 26 \times 5 = 130$ thousand; New salary = old salary + rise in total (1); $= 20 + 130 = 150$ thousand, i.e. ₹150000 (1).	1+1
12	Let amounts be x and y; $6x + 51y = 32(x + y)$ (1); $26x = 19y$; $x : y = 19 : 26$ (1).	1+1
13	Take distance each way $= 180$ km, so total distance $= 360$ km; Time $= \frac{180}{36} + \frac{180}{45} = 9$ h (1); Average speed $= \frac{360}{9} = 40$ km/h (not the simple mean of 36 and 45) (1).	1+1
14	Rise in total $= 22 \times 4 = 88$ thousand; New salary = old salary + rise in total (1); $= 36 + 88 = 124$ thousand, i.e. ₹124000 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $63 \times 92 + 81n = 88(63 + n)$ (1); $n(81 - 88) = 63(88 - 92)$ (1); $n = 36$ (1).	1+1+1
16	Weighted sum $= 96 \times 5 + 52 \times 7 + 74 \times 3 + 66 \times 4 = 1330$ (1); Sum of weights $= 19$ (1); Weighted average $= 70$ (1).	1+1+1
17	Original sum $= 570$; New sum $= 570 + 70 + 56 = 696$ (1); New mean $= \frac{696}{12} = 58$ (1); Rises (1).	1+1+1
	OR: Let the number of remaining students be n; $27 \times 54 + 87n = 78(27 + n)$ (1); $n(87 - 78) = 27(78 - 54)$ (1); $n = 72$ (1).	1+1+1
18	Weighted sum $= 90 \times 6 + 67 \times 2 + 56 \times 3 + 58 \times 6 = 1190$ (1); Sum of weights $= 17$ (1); Weighted average $= 70$ (1).	1+1+1
	OR: Original sum $= 558$; New sum $= 558 + 55 + 27 = 640$ (1); New mean $= \frac{640}{20} = 32$ (1); Rises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 + 3 + 5 = 10$ (1). (ii) $170 + 285 + 375 = 830$ (1). (iii) $\frac{830}{10} = 83$ (1). (iv) Simple mean $= \frac{85 + 95 + 75}{3} = 85$ (1); the Exam has the greatest weight and his lowest score, so the weighted average is pulled below the simple mean (1).	1+1+1+2
20	Cost of the mixture $= \frac{3 \times 65 + 1 \times 185}{4}$ (1); $= 95$ per kg (1); Profit factor $= \frac{120}{100}$ (1); Selling price $= 114$ per kg (2).	1+1+1+2
	OR: January food $= 2035$ (1); February food $= 1475$ (1); January is higher by 560 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2

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